

Bell Prover - Calibration and Monitoring of Time Stability

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Abstract

This paper is focused on the calibration and measurement of the geometric parameters of a bell prover that serves as a standard of volume of gas flow. It also includes methods for monitoring the time stability of the bell prover: intra- and inter-laboratory comparison measurements, Shewhart analysis and planned evaluation of recalibration intervals.

1. Introduction

The primary standard of gas flow - the bell prover - was established by the Slovak Institute of Metrology (SMU) during 1999 to 2002. The flow range of this standard is from 2 to 65 m³/h with an expanded uncertainty of 0.12 % [1,2]. This standard is listed in the Calibration and Measurement Capability tables of the BIPM (SK 13) [3].

2. Mathematic Model of a Bell Prover

The volumetric flow realized by a primary standard bell prover is evaluated as follows [4]:

$$Q_{VZ} = \frac{h_c \cdot S_c}{\Delta \tau} + \frac{\Delta \rho_a \cdot V_a}{\rho_c \cdot \Delta \tau} + Q_{Vs} =$$

$$\frac{\sum_{j=1}^n K_j \cdot I_j}{\Delta \tau} + \frac{\Delta \rho_a \cdot V_a}{\rho_c \cdot \Delta \tau} + Q_{Vn} \quad [m^3 \cdot h^{-1}] \quad (1)$$

where: h_c is the length of the vertical displacement of the bell [m]; S_c the cross-sectional area of the bell [m²]; V_a , the residual volume of air in the system [m³]; $\Delta \tau$, the time interval of measuring cycle [h]; ρ_c , the density of discharged air [kg. m⁻³]; $\Delta \rho_a$, the change in air density during the measuring cycle [kg. m⁻³]; Q_{Vn} , the volumetric flow through leaks [m³. h⁻¹]; K_j , the pulse coefficient [m³. imp⁻¹]; and I_j is the number of pulses [imp].

3. Calibration of the Bell Prover

Calibration includes the bell prover internal geometric dimensions at 110 horizontal levels. Furthermore, instruments installed in the primary standard were calibrated and their associated uncertainties were determined so they could be included in the uncertainty budget of Eq. (1).

3.1 Determination of Geometrical Parameters of the Bell

A special structure was manufactured for centering of a measuring device placed inside the primary standard frame and for attaching a measuring

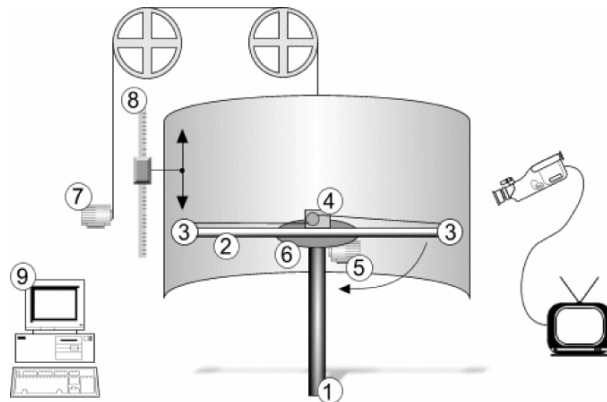


Fig. 1. Scheme of geometrical parameters bell measuring equipment

arm (Fig. 1, position 1). The platform of this structure (position 6), where the arm was clamped, was equipped with a stepper motor (position 5) for 180° horizontal displacement. A photograph of the apparatus for the measurement of the geometrical parameters of the bell is shown in Fig. 2.

After preliminary tests, the contact method was selected: digital micrometers with a communication interface (position 3), measuring range of 12 mm, resolution of 1 µm, and the possibility of remote transmission of measured data were used. Unequal circumferential separation of measuring points was specified based on the final bell manufacturing. Points separated by 50 mm distance were generally designated on the circumference of the inner casing but 10 mm distance was used around the weld. Measurements in the horizontal plane were made every 10 mm over the 1100 mm working stroke. A vertical displacement was performed by bell displacement, which was driven using a stepper motor for the vertical ruler movement (position 7). Location of the bell was taken using an optical ruler (position 8). Shifting in the vertical direction was carried out when the micrometer tips were out of contact with the bell (to protect the tips against abrasion). This was ensured by an assembly mechanism (position 4), that detached tips from the bell after the reading. The process of determining geometrical parameters runs fully automatically and was controlled from the control centre by a computer program designed specifically for these



Fig. 2. Apparatus for measurement of the geometrical parameters of the bell

measurements (position 9). The measurement was performed in an air-conditioned enclosure at $(22 \pm 0.1) ^\circ\text{C}$. A video camera system was used to monitor the measurements inside the bell and the video was recorded in the control center. The measurements were performed at fifty diameters in each of the 110 horizontal planes resulting in 5 500 measurement points. The time interval of one cycle of measurements was approximately 12 hours. Every cycle was repeated three times and data were recorded in the database files [4].

The diameter was calculated according to the following:

$$d_{ij} = 1099.99 + 12.0 a_{1,ji} - a_{2,ji} [\text{mm}] \quad (2)$$

where: d_{ij} is the measured internal diameter of the bell at an angle j , j on the height h_j [mm]; $a_{1,ji}$ is the value given by the sensor No. 1 [mm]; $a_{2,ji}$ is the value

given by the sensor No. 2 [mm]; the value of 1099.99 is the statement from the calibration certificate of the measuring arm [mm] and the value of 12.0 is the sum of half-strokes of both micrometers, on which micrometers by calibration of measuring arm were set [mm].

These measured data of the inner diameters of the bell were then transferred to the computer program where the area for each plane (Fig. 3) was calculated using the following equation:

$$s_j = \sum_i (p_{1,j,i} + p_{2,j,i}) [\text{mm}^2] \quad (3)$$

where s_j is the surface of the bell in the plane j ; $p_{1,j,i}$ is the surface of the bell in the plane j , in the segment i , for the angle $(\alpha_{j,i+1} - \alpha_{j,i})$, for the sensor No. 1 [mm²] and $p_{2,j,i}$ is the a surface of the bell in the plane j , in the segment i , for the angle $(\alpha_{j,i+1} - \alpha_{j,i})$, for the sensor No. 2 [mm²].

Area of the segment $p_{1,j,i}$ is expressed in the following equation:

$$p_{1,j,i} = \frac{\alpha_{j,i+1} - \alpha_{j,i}}{6} \cdot \left(r_{1,j,i+1}^2 + r_{1,j,i+1} r_{1,j,i} + r_{1,j,i}^2 \right) [\text{mm}^2] \quad (4)$$

where:

$$r_{1,j,i} = \frac{1099.99}{2} + 6 - \alpha_{1,j,i} \quad (5)$$

is a radius determined by sensor No. 1 in the plane j by measurement i .

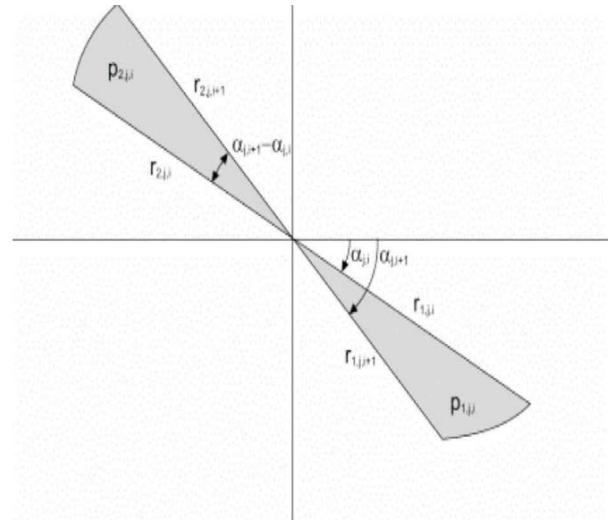


Fig. 3. Graphical representation of measurement of the area of the bell

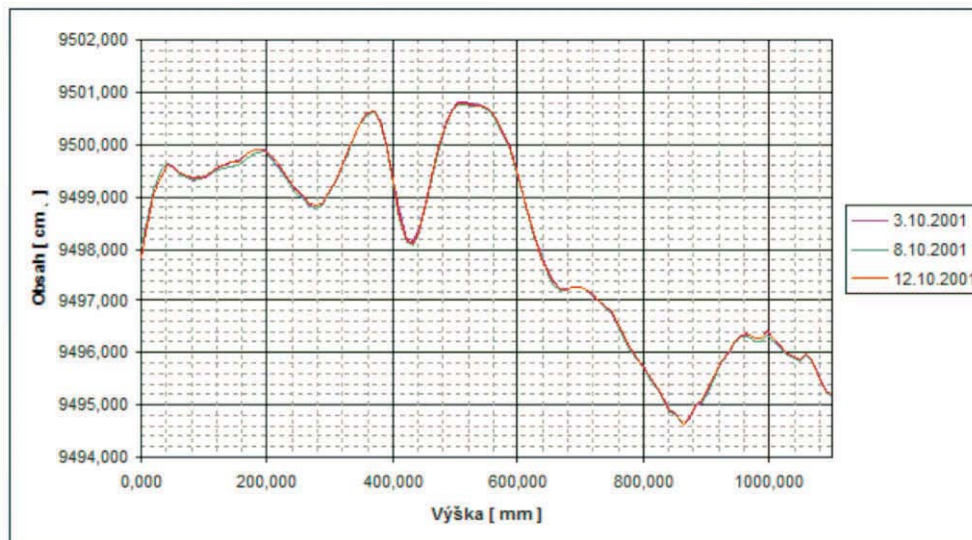


Fig. 4. Charts of the inner bell diameter for the 110 individual horizontal planes

Similarly, the area of the segment $p_{2,j,i}$ is expressed in the following equation:

$$P_{2,j,i} = \frac{\alpha_{j,i+1} - \alpha_{j,i}}{6} \cdot (r_{2,j,i+1}^2 + r_{2,j,i+1}r_{2,j,i} + r_{2,j,i}^2) [\text{mm}^2] \quad (6)$$

where:

$$r_{2,j,i} = \frac{1099.99}{2} + 6 - a_{2,j,i} [\text{mm}] \quad (7)$$

is the given radius determined by the sensor No. 2 in the plane j by measurement i .

Areas of the bell in each of the cross sections were calculated from the measured values and the calculation entered into the database of the evaluation program of the bell prover. The set of calculated values of the inner diameter of the bell in the 110 horizontal planes, measured on three occasions, is presented in Fig. 4.

3.2 Sources of Uncertainty of Flow Measurement of the Bell Prover

Table 1
Uncertainty budget of the bell prover

Source of uncertainty	Relative standard uncertainty [%]
Collection volume	0.015
Bell area	0.007
Collection length	0.008
Oil film adherence	0.010
Air density	0.030
Temperature	0.020
Pressure	0.010
Compressibility	0.020
Collection time	0.040
Timer calibration	0.001
Timing error	0.040
Residual volume	0.010
Leakages	0.010
Relative standard uncertainty of flow	0.058
Relative expanded uncertainty of flow	0.116

4. Inter-laboratory Comparisons

The primary standard was bilaterally compared with the primary standard of PTB Braunschweig, Germany in 2002-2005[5]. The bell prover was declared in 2007 as a national standard of gas flow in the Slovak Republic [6].

The Euramet and CIPM Key comparisons of national (primary) standards of gas flow were initiated by the Slovak Institute of Metrology and they were prepared with the co-operation of PTB for 2010-2011. The results of these comparisons will be used by SMU for independently checking the bell prover main uncertainty contributions [7].

5. Time Stability Monitoring

Further it was necessary to have reliable information about the state of standard equipment in the period between international comparisons. Therefore, the Centre of Flow of Slovak Institute of Metrology developed a method for regularly monitoring the bell prover measuring system time stability. The method is based on collecting a data set that can be used for monitoring (intra-laboratory comparisons, a set of control charts and planned evaluation of recalibration intervals). Fixed points of flow - the critical nozzles - were used as a witness standard for the measuring range of flow, too.

5.1 Intra-Laboratory Comparisons

Eight intra-laboratory measurement comparisons were performed during the years 2005 to 2009 between the bell prover and the standard with the equivalent amount of liquid. The standard with the equivalent amount of liquid [2] is a part of the national standard of gas flow. Results of intra-laboratory comparisons are given in the Fig. 5. Values of standardized deviation E_n did not exceed the value of $|0,4|$.

5.2 Shewhart Analysis

Shewhart regulatory diagrams [8] were chosen as a second tool for the bell prover time stability monitoring. These diagrams are used for identification of determinable causes of variability in the measurement process during production. The measurements were provided by calibration of the critical nozzles (witness standards) with the bell prover (Figs. 6-7).

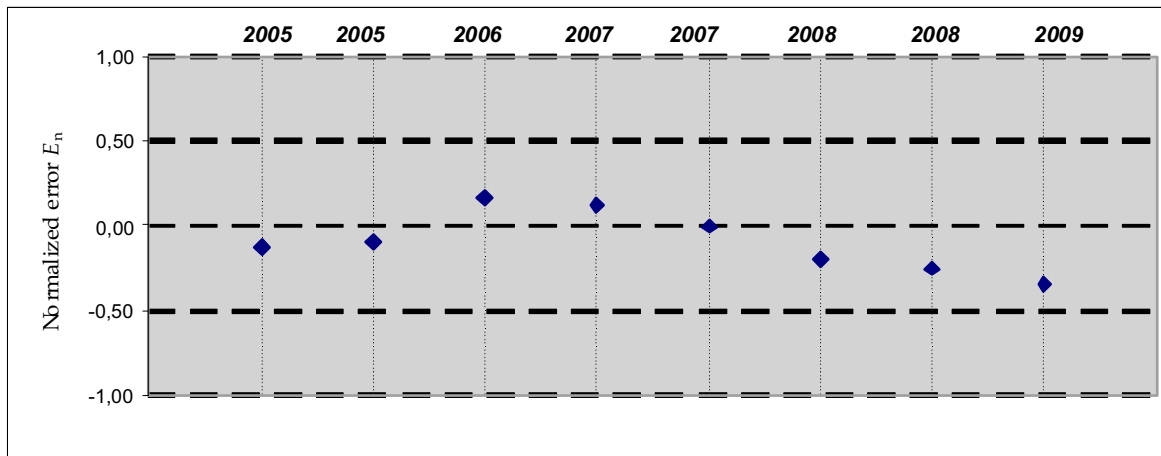


Fig. 5. Normalized errors of intra-laboratory comparisons from 2005 to 2009 between the bell prover and the standard with the equivalent volume of liquid

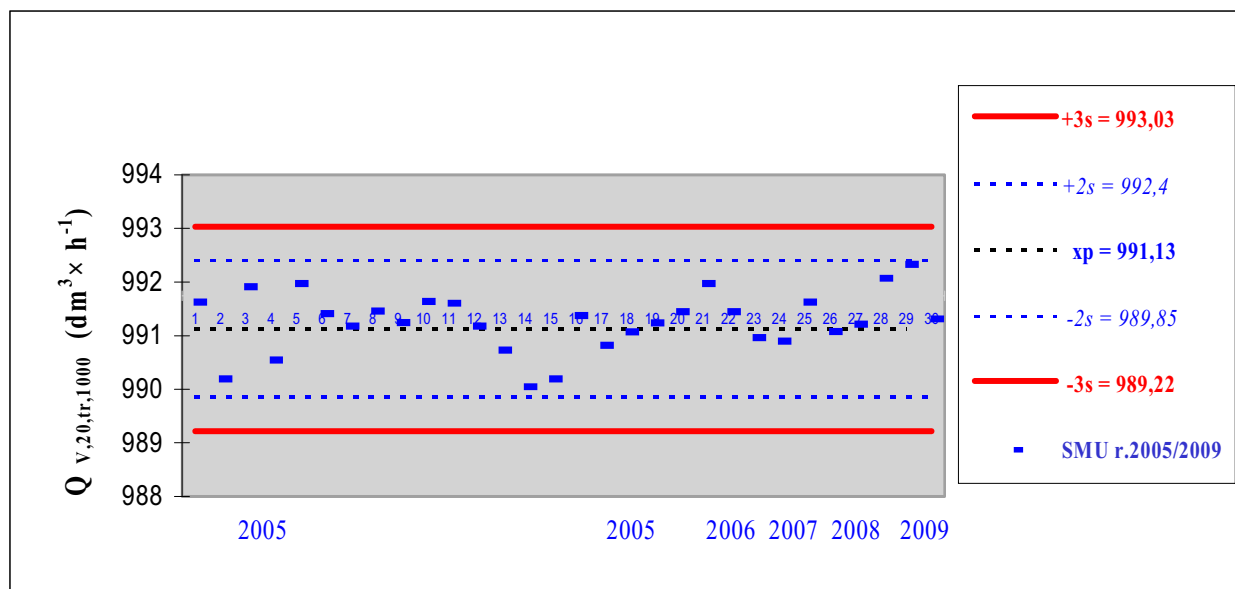


Fig. 6. Regulatory diagram with critical nozzle JT 01-1000-99 in the lower flow range ($1 \text{ m}^3/\text{h}$) - from year 2005 to 2009

The diagrams show the central line representing the arithmetic mean of flow of the bell prover. Other parallel lines to the central line are warning limits ($2s$) and control limits ($3s$), where s is the standard deviation of measurement for a critical nozzle. All points measured in 2005 - 2009 lie within the warning limits and therefore the time stability of the bell prover is found to be statistically stable.

5.3 Planned Project of Recalibration Intervals Evaluation

The Centre of Flow has launched a project for recalibration interval determination for flow standards. Special attention is paid to the gas flow national standard, and the 2001 measurement of the bell internal dimensions. Monitoring the time stability

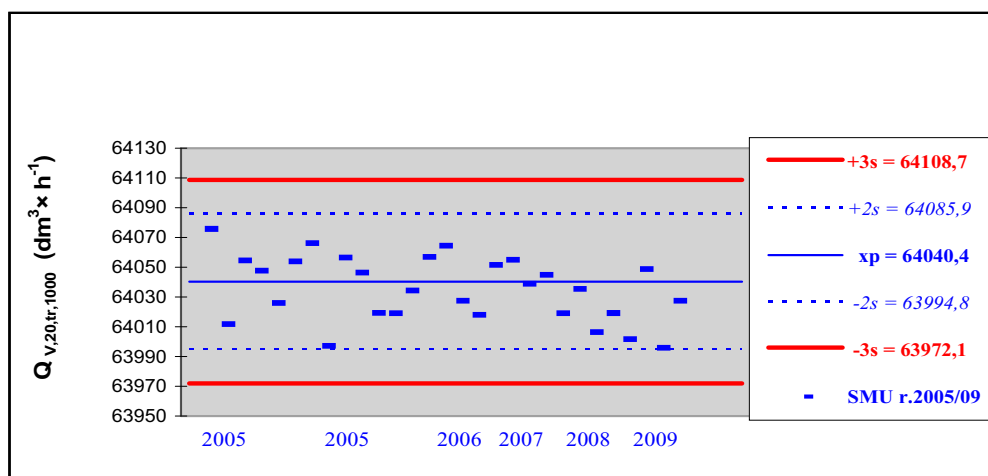


Fig. 7. Regulatory diagram with critical nozzle 02-64000-99 in the upper range of flow ($64 \text{ m}^3/\text{h}$) - from year 2005 to 2009

of the bell prover will continue. However, analysis and design of the recalibration interval will determine when it is necessary for the next measurement of the

geometrical parameters of the bell. The project is based on the document OIML and NASA Guide [9, 10].

6. Results

Table 2
Uncertainty budget of the bell prover

Monitoring tool	Criteria	Fulfill of criteria	Note
Intra-laboratory comparison	$ E_n < 1$	$-0.4 < E_n < 0.2$	-
Shewhart analysis	$x_p - 3s < Q < x_p + 3s$	$x_p - 2s < Q < x_p + 2s$	-
Recalibration intervals	planned	-	project in year 2011

7. Conclusion

Previous periodic monitoring in 2005-2009 exhibited that the standard maintained stability within specified limits. The methods described in sections 4.1 and 4.2 consistently confirm the time stability of bell prover.

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